Overall Survival Prediction in Adult Myeloid Leukemia: Kaplan–Meier, Cox Regression and Machine-Learning Survival Models

Bernard Frimpong
M.S. Applied Mathematics, University of Akron

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Abstract

This report documents my work on the QRT × Institut Gustave Roussy data challenge on Challenge Data (ENS Paris), whose goal is to rank adult patients with myeloid leukemia by their risk of death, using clinical measurements, cytogenetic reports and somatic gene mutations. Models are scored with the inverse probability of censoring weighted (IPCW) concordance index truncated at seven years. The data audit shows that the 1,193 test patients all come from a single clinical centre that is absent from the 23 training centres, and that this centre's patients are clinically sicker, so model validation will hold out whole centres. Kaplan–Meier analysis gives a median overall survival of 3.15 years, with strong separation by bone-marrow blasts, *TP53* and *SF3B1* mutations and monosomy 7 (log-rank $p < 0.001$ for all). The Kaplan–Meier estimator, a log-rank test and both concordance indices were computed by hand and match standard software. Single raw variables reach an IPCW C-index of at most 0.616. I then built 91 features from blood counts, a forgiving parser of the free-text karyotypes, and somatic mutations restricted to the 64 genes sequenced in both centres. The published IPSS-R clinical risk score, which needs no fitting, reaches an IPCW C-index of 0.669 ± 0.029 across 15 centre-grouped validation folds. Cox models evaluated on the same folds improve on it: a 24-variable model built from the IPSS-M gene list reaches 0.716 and an elastic-net Cox model on 84 features 0.722, each beating the IPSS-R on all 15 folds. Monocyte counts, missing in 74% of test patients, add nothing. Hazard ratios agree with clinical knowledge (TP53 multi-hit HR 3.2), and the proportional-hazards assumption is violated mainly by *NPM1* and platelets, whose effects change over time. On the public leaderboard the elastic-net scores 0.7543. Tree models (a random survival forest and XGBoost with a Cox loss, tuned by nested cross-validation) do not improve on it: XGBoost ties the elastic-net at 0.723 and the forest reaches 0.719, suggesting the engineered features already capture most of the structure the trees could find.

1 Introduction

Myeloid leukemias are blood cancers in which abnormal immature cells (blasts) accumulate in the bone marrow. Estimating a patient's prognosis guides treatment: low-risk patients may receive supportive care, while high-risk patients may be prioritised for intensive treatment such as stem cell transplantation.

The outcome is *overall survival* (OS): the time from diagnosis to death or to the last follow-up. When a patient is still alive at the last follow-up, the true survival time is unknown; we only know it is longer than the observed time. Such observations are called *right-censored*, and handling them correctly is the purpose of survival analysis.

The project follows a staged plan: Kaplan–Meier estimation and log-rank tests, Cox proportional hazards regression, and machine-learning survival models (gradient-boosted survival and random survival forests), all compared under the challenge metric with honest cross-validation.

A similar crowdsourced effort, the DREAM 9 AML Outcome Prediction Challenge, also scored survival models with a concordance index [1].

2 Data

The data come from 24 clinical centres and are split into the five files described in Table 1. The `ID` column links the files.

Table 1: The five challenge files.

File	Rows	Content
`clinical_train.csv`	3,323	one row per patient: centre, blood counts, karyotype
`clinical_test.csv`	1,193	same, for test patients
`molecular_train.csv`	10,935	one row per somatic mutation
`molecular_test.csv`	3,089	same, for test patients
`target_train.csv`	3,323	`OS_YEARS`, `OS_STATUS`

Clinical variables. `BM_BLAST` (bone-marrow blasts, %), `WBC` (white blood cells, G/L), `ANC` (absolute neutrophil count, G/L), `MONOCYTES` (G/L), `HB` (haemoglobin, g/dL), `PLT` (platelets, G/L) and `CYTOGENETICS`, a free-text karyotype written in ISCN notation (for example `46,xy` for a normal male karyotype).

Molecular variables. Each row is one somatic mutation, with its chromosomal position, reference and alternate bases, the gene, the protein change, the broad effect, the variant allele fraction (`VAF`, the share of sequencing reads carrying the mutation) and, not mentioned in the challenge description, the sequencing read depth (`DEPTH`).

3 The evaluation metric

For each patient i we observe a time T_i (`OS_YEARS`), an event indicator Δ_i (1 = death, 0 = censored), and the model gives a risk score η_i.

Harrell's C-index. A pair of patients (i, j) is *comparable* when we know who died first, that is, when $T_i < T_j$ and patient i died ($\Delta_i = 1$). The pair is *concordant* when the model agrees, $\eta_i > \eta_j$. Harrell's C-index [2] is

$$C = \frac{\text{number of concordant comparable pairs}}{\text{number of comparable pairs}}. \tag{1}$$

A value of 0.5 corresponds to random guessing and 1 to a perfect ranking. Pairs with equal risk scores count as one half.

IPCW C-index (Uno's C). Patients who die late are less likely to be observed dying, because many patients are censored before then. Harrell's C therefore over-represents early deaths. Uno *et al.* [3] correct this by weighting each comparable pair by $1/\hat{G}(T_i)^2$, where $\hat{G}(t)$ is the Kaplan–Meier estimate of the probability of *not yet being censored* at time t. With the truncation horizon $\tau = 7$ years used by the challenge,

$$C_\tau = \frac{\displaystyle\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)\, \mathbf{1}(\eta_i > \eta_j)}{\displaystyle\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)}. \tag{2}$$

In plain language: a death at a time when few patients remain uncensored is rare to observe, so it counts more. Only the ordering of the scores η_i matters. The challenge computes C_τ with `scikit-survival` [4].

4 Data audit (Stage 1)

4.1 Linking and outcome

All IDs are unique, every mutation row belongs to a known patient, and no patient appears in both training and test sets. However, 150 training patients have no recorded outcome and are excluded from modelling. Table 2 summarises the remaining outcomes and Figure 1 shows the observed times.

Table 2: Outcome of the training patients.

Quantity	Value
Training patients	3,323
Without outcome (excluded)	150
With outcome	3,173
Deaths ($\Delta = 1$)	1,600 (50.4%)
Censored ($\Delta = 0$)	1,573 (49.6%)
Median observed time (years)	1.65
Maximum observed time (years)	22.04
Observed time exactly 0 (all censored)	53
Observed beyond the 7-year horizon	172

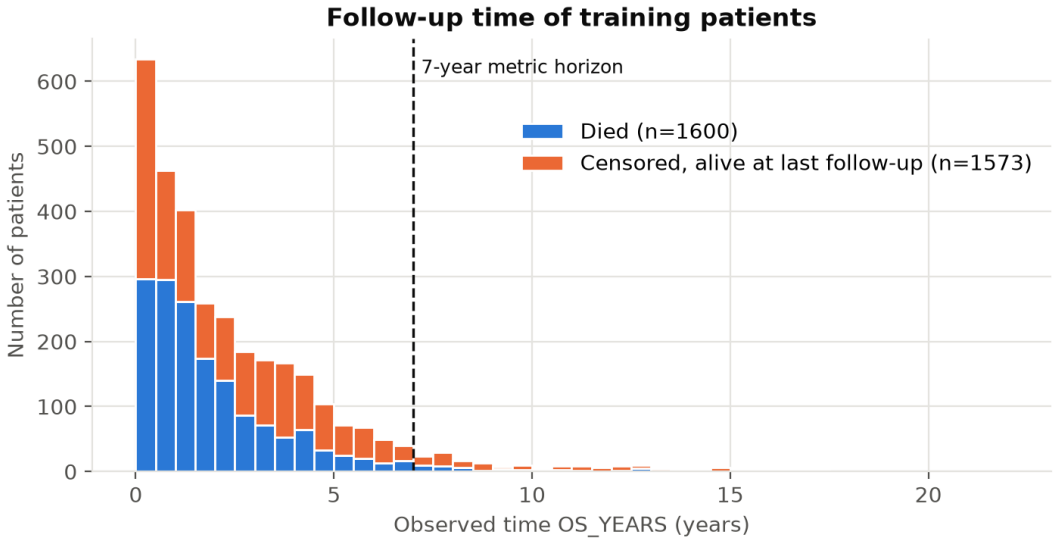


Figure 1: Observed time of the 3,173 training patients with an outcome, split into deaths and censored patients. The dashed line marks the 7-year horizon of the metric.

About half the patients are censored, so simply discarding them or treating their observed time as a death time would badly bias any model. Most deaths occur within the first two years, and only 172 patients are followed beyond seven years, which justifies the truncation of the metric.

4.2 Clinical centres: the test set is a new hospital

The 3,323 training patients come from 23 centres, while all 1,193 test patients come from one centre, `KYW`, that does not appear in training (Figure 2). Death rates also vary widely between training centres, from 11% (`ROM`) to 92% (`MUV`), reflecting different patient populations and follow-up lengths.

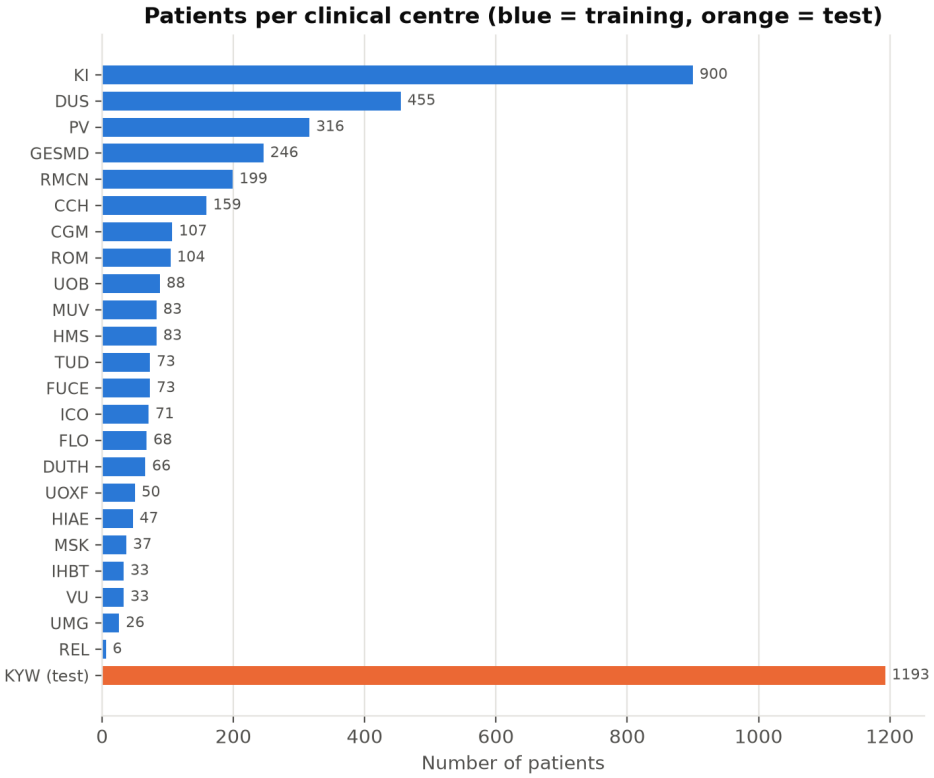


Figure 2: Number of patients per clinical centre. The test centre `KYW` (orange) contributes no training patients.

This has a direct consequence for validation. A random K-fold split mixes patients from the same centre into training and validation folds, so the model is always validated on centres it has already seen. The real task is to predict for an *unseen* centre. Cross-validation will therefore be grouped by centre, and the `CENTER` variable cannot be used as a predictor.

4.3 Clinical variables: training versus test

Table 3 and Figures 3 and 4 compare the clinical variables in the two sets.

Table 3: Missing values and medians of the clinical variables.

	Missing (%)		Median	
Variable	Train	Test	Train	Test
`BM_BLAST` (%)	3.3	9.6	3.00	6.00
`WBC` (G/L)	8.2	9.4	4.10	2.90
`ANC` (G/L)	5.8	11.8	2.00	1.30
`MONOCYTES` (G/L)	18.1	74.0	0.37	0.20
`HB` (g/dL)	3.3	9.3	9.70	8.40
`PLT` (G/L)	3.7	9.6	123.00	74.00
`CYTOGENETICS`	11.6	9.7	–	–

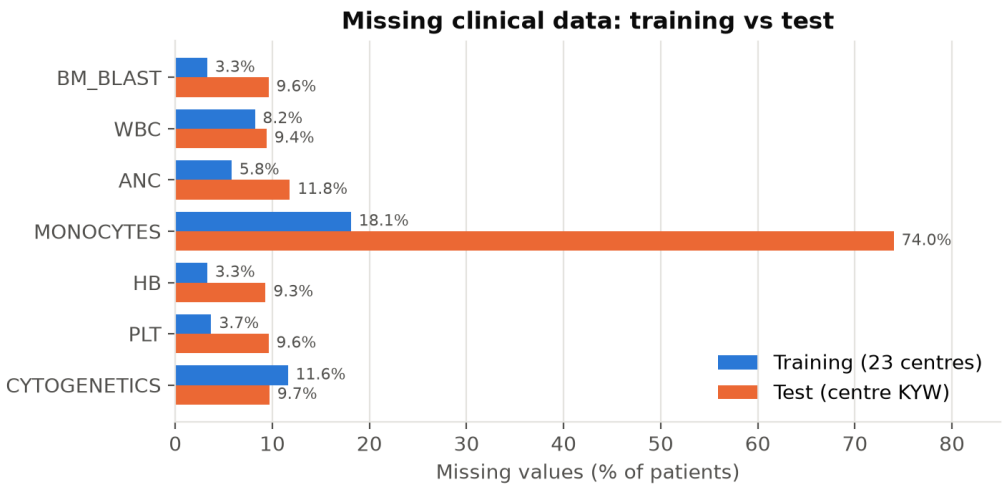


Figure 3: Percentage of missing values in each clinical variable.

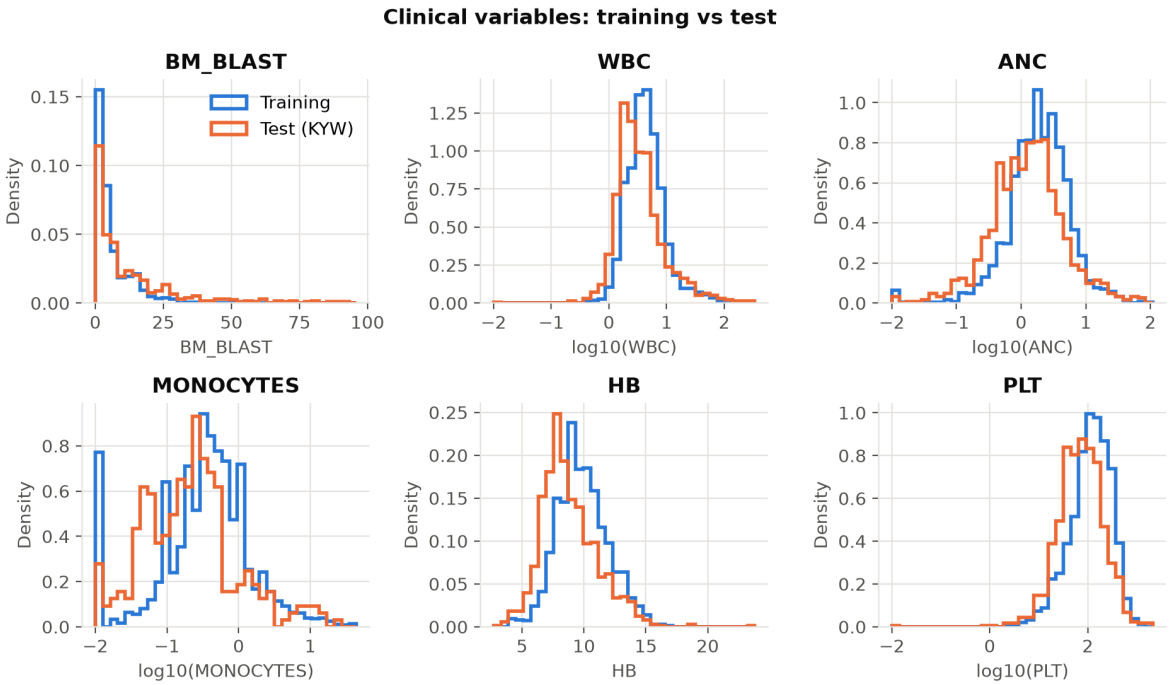


Figure 4: Distributions of the clinical variables in training (blue) and test (orange). Each histogram is scaled to area one so that sets of different sizes can be compared. Skewed blood counts are shown on a $\log_{10}$ scale; the spike at -2 for monocytes corresponds to recorded values of zero.

The test patients are clinically sicker: lower haemoglobin and platelets, lower neutrophils, and a much larger share with bone-marrow blasts above 20% (17.6% versus 4.2%). Monocytes are missing for 74% of test patients (883 of 1,193), so a model that leans on that variable would be fragile on the test set: all 883 would receive the same imputed value, and the training median (0.37) is above the test centre's own median (0.20).

4.4 Gene mutations

Training patients carry on average 3.29 recorded mutations and test patients 2.59; 297 training and 139 test patients have none, which is genuine information rather than missing data. Training data contain 124 distinct genes and test data 83, of which 19 never appear in training. Of the training rows, 114 lack a chromosomal position: they are partial tandem duplications of *MLL* (89) and internal tandem duplications of *FLT3* (25), which are structural events rather than single-base changes. In 1,274 cases the same gene is mutated more than once in one patient.

Table 4: Ten most frequently mutated genes in training: percentage of patients with at least one mutation.

Gene	Train (%)	Test (%)
TET2	31.1	17.5
ASXL1	26.9	12.7
SF3B1	22.2	7.7
SRSF2	17.2	9.1
DNMT3A	16.1	11.1
RUNX1	13.9	13.7
TP53	11.4	25.7
STAG2	9.1	6.7
U2AF1	8.6	8.4
EZH2	6.5	6.7

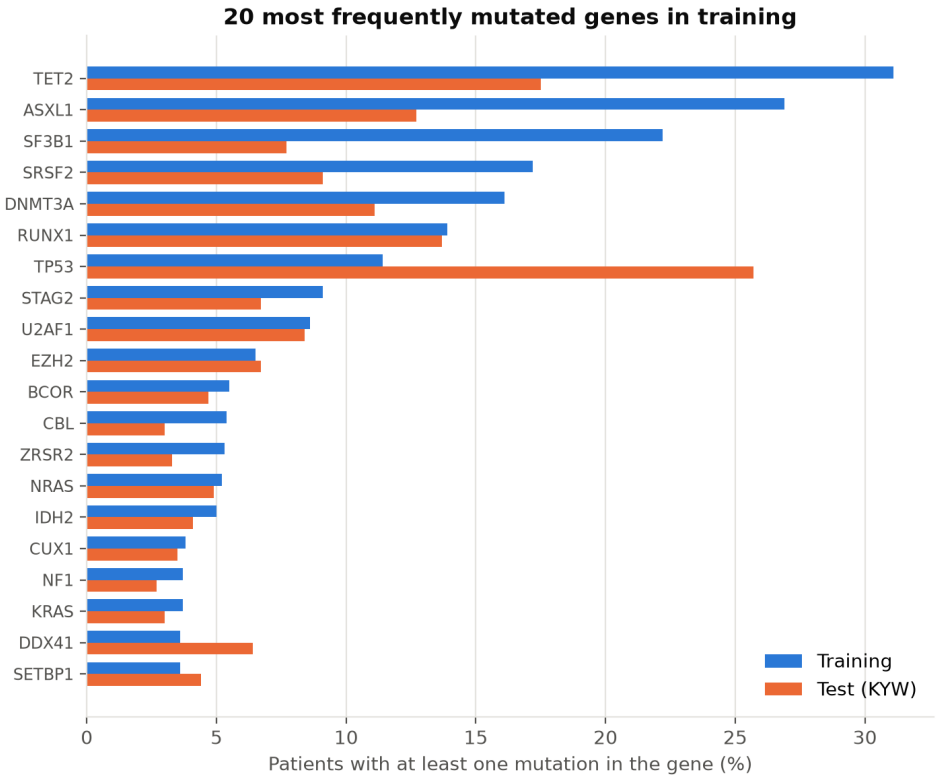


Figure 5: The twenty most frequently mutated genes in training, with their frequency in the test centre.

The most striking difference is *TP53*, mutated in 25.7% of test patients against 11.4% in training (Table 4, Figure 5). *TP53* mutations are associated with poor prognosis in myeloid neoplasms, which is consistent with the sicker clinical profile of the test centre. By contrast, *SF3B1* mutations, typically associated with lower-risk disease, are almost three times rarer in the test set.

4.5 Cytogenetics text

In training, karyotypes are written in lower case (`46,xy`); in the test set they are written in upper case (`46,XY`), and 241 test patients have the single word `Normal`. Any parsing must first convert

text to lower case and map `Normal` to a normal karyotype; otherwise every test karyotype would look unfamiliar to the model.

4.6 Consequences for modelling

1. Exclude the 150 training patients without outcome.
2. Validate with cross-validation grouped by centre, and do not use `CENTER` as a predictor.
3. Prefer features that are measured consistently in both sets; treat `MONOCYTES` with caution.
4. Normalise the cytogenetics text before extracting features.
5. Encode "no mutation found" as zero, not as missing.

5 Survival analysis without a model (Stage 2)

5.1 The Kaplan–Meier estimator

Let $t_1 < t_2 < \dots$ be the distinct times at which at least one death occurs. At each t_k let n_k be the number of patients still *at risk* just before t_k (neither dead nor censored yet) and d_k the number of deaths at t_k. The Kaplan–Meier estimator [5] of the survival function is

$$\hat{S}(t) = \prod_{k:\ t_k \le t} \left(1 - \frac{d_k}{n_k}\right). \tag{3}$$

In plain language: to survive past t_k you must survive every earlier death time, and at each one the chance of surviving, given you were still alive, is the share of the at-risk patients who did not die. Censored patients never make the curve drop; they simply leave the risk set, so they count fully for as long as they were observed.

Table 5 shows the first six steps. The first death occurs on day one ($t = 0.00274$ years) with $n = 3{,}120$ patients at risk: this is $3{,}173 - 53$, because the 53 patients with an observed time of zero were all censored at that time and left the risk set immediately. The first two steps are

$$\hat{S}(0.00274) = 1 - \tfrac{1}{3120} = 0.99968,$$
$$\hat{S}(0.00822) = 0.99968 \times \left(1 - \tfrac{2}{3114}\right) = 0.99968 \times 0.99936 = 0.99904.$$

My hand implementation of Equation (3) agrees with the `lifelines` library [7] at every death time to within 2×10^{-14}.

Table 5: First six steps of the Kaplan–Meier estimator, computed by hand.

t_k (years)	n_k	d_k	$1 - d_k/n_k$	$\hat{S}(t_k)$
0.00274	3120	1	0.99968	0.99968
0.00822	3114	2	0.99936	0.99904
0.01644	3107	1	0.99968	0.99872
0.02192	3103	2	0.99936	0.99807
0.02466	3098	1	0.99968	0.99775
0.02740	3095	2	0.99935	0.99711

The median overall survival, the first time at which $\hat{S}(t) \le 0.5$, is **3.15 years**. Table 6 and Figure 6 give the whole curve.

Table 6: Kaplan–Meier overall survival with 95% confidence intervals.

Years	$\hat{S}(t)$	95% CI
1	0.790	0.774–0.804
2	0.614	0.595–0.633
3	0.509	0.489–0.529
5	0.364	0.342–0.386
7	0.275	0.250–0.299

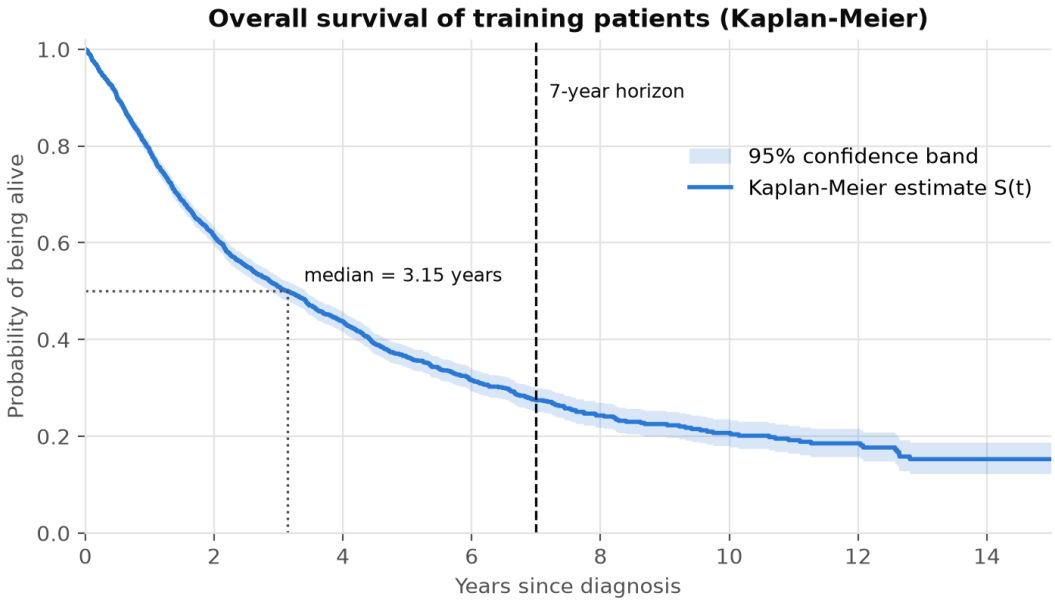


Figure 6: Kaplan–Meier estimate of overall survival for the 3,173 training patients, with a 95% confidence band.

5.2 Comparing groups: the log-rank test

To ask whether two groups have different survival, the log-rank test [6] compares, at each death time, the deaths observed in group 1 with the deaths it would receive if the groups were identical. With n at risk in total, n_1 in group 1, $n_2 = n - n_1$, d deaths in total and d_1 in group 1,

$$E_{1k} = d\,\frac{n_1}{n}, \qquad V_k = \frac{n_1\, n_2\, d\,(n-d)}{n^2(n-1)}, \qquad \chi^2 = \frac{\left(\sum_k (d_{1k} - E_{1k})\right)^2}{\sum_k V_k}, \tag{4}$$

which follows a χ^2 distribution with one degree of freedom when the groups do not differ. In plain language: group 1 should get "its fair share" n_1/n of the deaths at each time; χ^2 measures how far the total gap is from zero, in units of its own variability.

Worked example: *TP53*. Summing over all death times, the 368 patients with a *TP53* mutation had $O_1 = 271$ observed deaths against $E_1 = 112.13$ expected, with total variance $V = 103.17$. Hence

$$\chi^2 = \frac{(271 - 112.13)^2}{103.17} = \frac{158.87^2}{103.17} = 244.64, \qquad p = 3.8 \times 10^{-55},$$

identical to the `lifelines` result. *TP53*-mutated patients died more than twice as often as expected under "no difference".

Table 7 and Figure 7 show four comparisons. Monosomy 7 was detected in the karyotype text as "-7" after a comma, slash or the start of the string and not followed by another digit; 79 patients without a blast value are left out of the blast comparison.

Table 7: Kaplan–Meier summaries and log-rank tests by group. All $p < 0.001$.

Comparison	Group	n	Deaths	Median (y)	$\hat{S}(3)$	χ^2 (df)
Blasts	$< 5\%$	2015	841	4.47	0.620	
	5–20%	942	622	1.68	0.326	272.1 (2)
	$> 20\%$	137	89	1.19	0.238	
TP53	not mutated	2805	1329	3.59	0.549	244.6 (1)
	mutated	368	271	1.00	0.207	
Monosomy 7	no	3008	1477	3.41	0.528	139.5 (1)
	yes	165	123	0.95	0.169	
SF3B1	not mutated	2446	1289	2.57	0.463	51.8 (1)
	mutated	727	311	4.56	0.665	

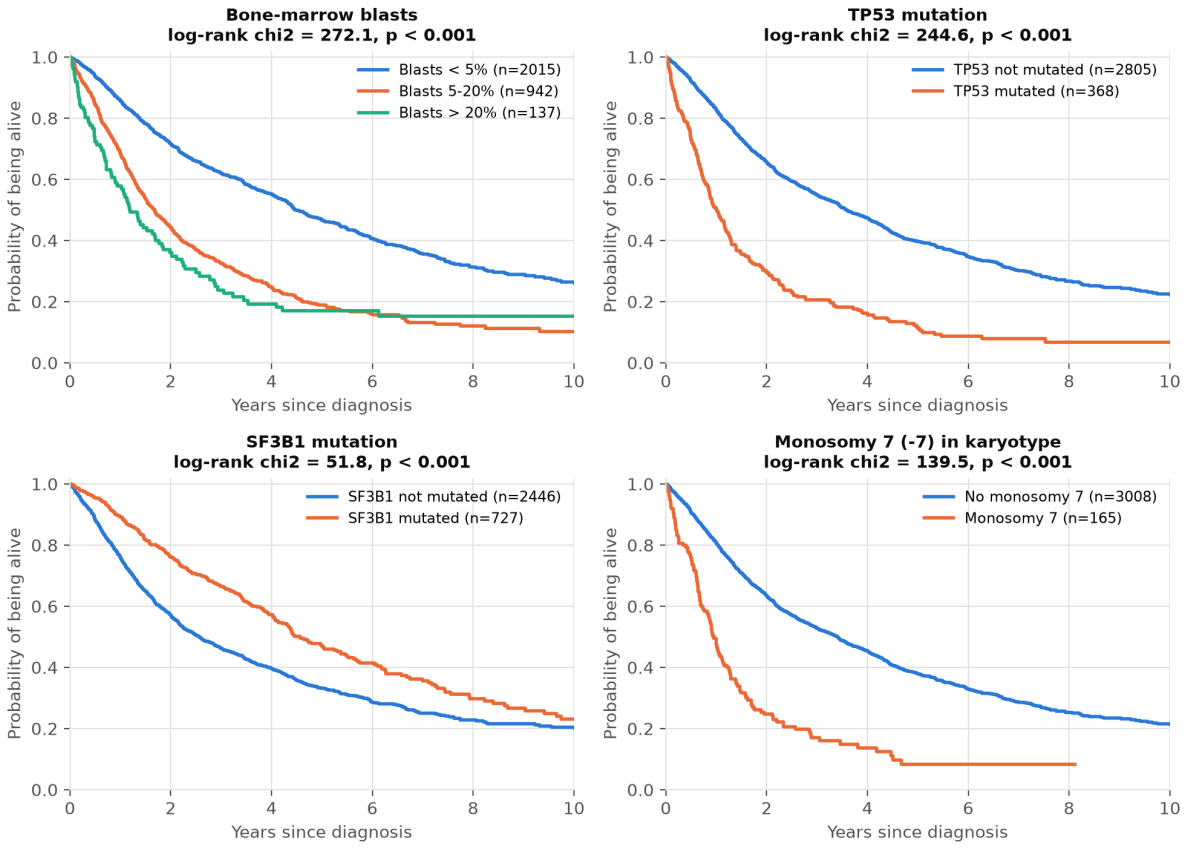


Figure 7: Kaplan–Meier curves by bone-marrow blast level, *TP53* mutation, *SF3B1* mutation and monosomy 7.

Higher blasts, *TP53* mutation and monosomy 7 all shorten survival sharply; the median survival of *TP53*-mutated patients is one year against 3.59 years for the rest. *SF3B1* is protective: mutated patients have a median survival of 4.56 years against 2.57. These directions agree with established clinical risk stratification and with the sicker profile of the test centre noted in

Stage 1.

5.3 The censoring curve

Swapping the roles of death and censoring in Equation (3), so that the "event" is being censored, gives the *reverse Kaplan–Meier* estimate $\hat{G}(t)$: the probability that a patient is still being followed at time t. This is exactly the $\hat{G}$ in the metric (Equation (2)).

Table 8: Censoring curve and the resulting metric weight of a death at time t.

t (years)	$\hat{G}(t)$	weight $1/\hat{G}(t)^2$
1	0.827	$1/0.827^2 = 1.46$
3	0.613	$1/0.613^2 = 2.66$
5	0.347	$1/0.347^2 = 8.32$
7	0.198	$1/0.198^2 = 25.61$

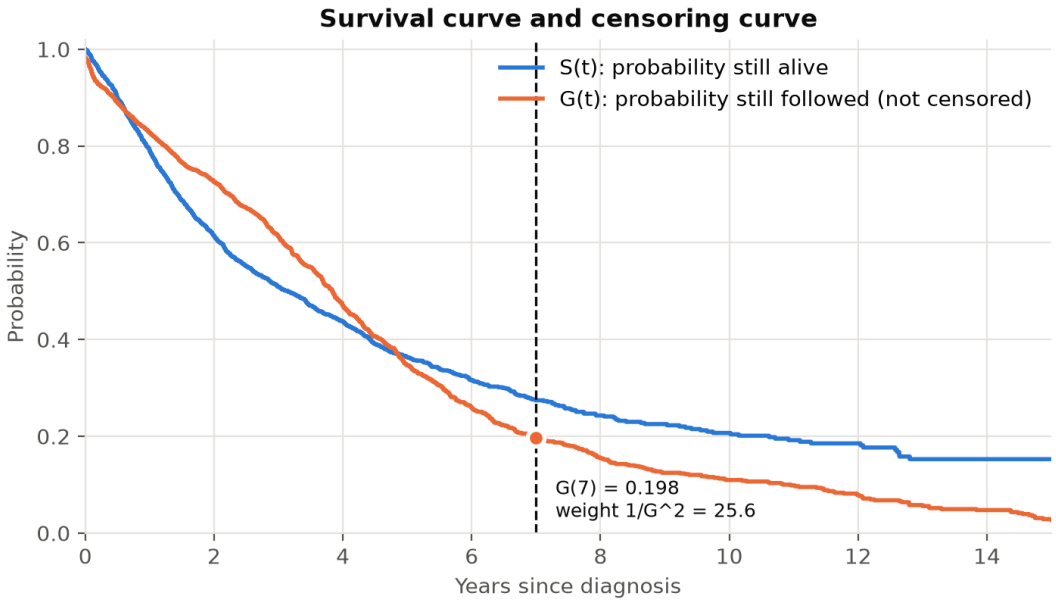


Figure 8: Survival curve $\hat{S}(t)$ and censoring curve $\hat{G}(t)$. By seven years only 19.8% of patients are still being followed.

Table 8 shows the practical effect: a pair started by a death just before seven years weighs about $25.6/1.46 \approx 17.5$ times as much as one started by a death at one year. A model that ranks late deaths poorly is therefore penalised far more by the challenge metric than by Harrell's C.

5.4 The metric by hand: a six-patient example

Figure 9 shows six patients with observed times, statuses and risk scores.

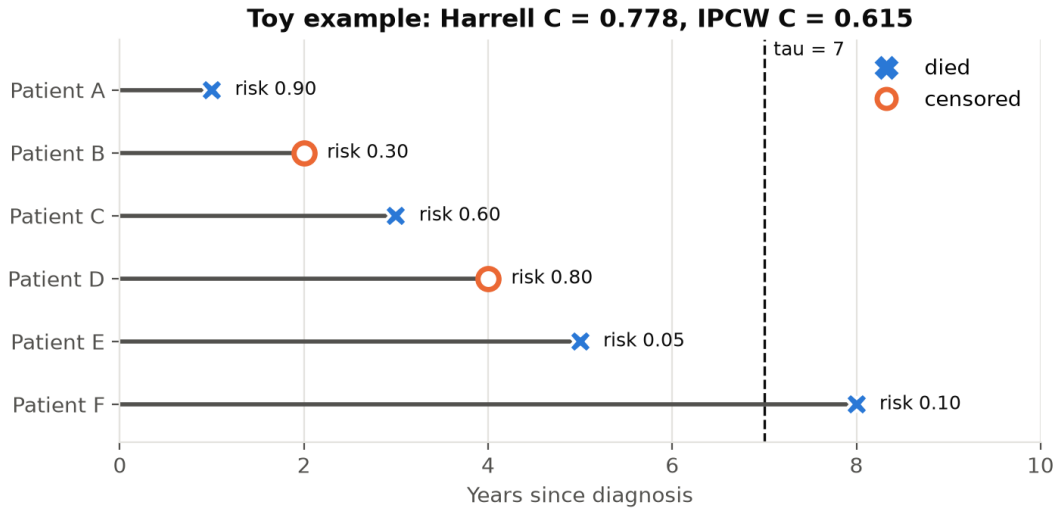


Figure 9: Toy example: crosses are deaths, circles are censored patients; each patient carries a risk score.

Comparable pairs. Only deaths can start a comparable pair, and only with patients observed for longer. Patient A (death at 1) is comparable with B, C, D, E, F; patient C (death at 3) with D, E, F; patient E (death at 5) with F; patient F (death at 8) has nobody after it. That gives $5 + 3 + 1 = 9$ comparable pairs. Of these, two are discordant: (C, D) because $0.60 < 0.80$ and (E, F) because $0.05 < 0.10$. So

$$C_{\text{Harrell}} = \frac{7}{9} = 0.778.$$

Censoring curve. Censorings occur at $t = 2$ (B), with 5 patients at risk, and at $t = 4$ (D), with 3 at risk:

$$\hat{G}(2) = 1 - \tfrac{1}{5} = 0.8, \qquad \hat{G}(4) = 0.8 \times \left(1 - \tfrac{1}{3}\right) = 0.5333.$$

So $\hat{G}(1) = 1$, $\hat{G}(3) = 0.8$ and $\hat{G}(5) = 0.5333$, giving the weights

$$w_A = \frac{1}{1^2} = 1, \qquad w_C = \frac{1}{0.8^2} = 1.5625, \qquad w_E = \frac{1}{0.5333^2} = 3.5156.$$

Patient F died at $8 \geq \tau = 7$ and starts no pair anyway.

IPCW C-index.

$$\begin{aligned} \text{numerator} &= 5 \times 1 + 2 \times 1.5625 + 0 \times 3.5156 = 8.1250, \\ \text{denominator} &= 5 \times 1 + 3 \times 1.5625 + 1 \times 3.5156 = 13.2031, \\ C_7 &= \frac{8.1250}{13.2031} = 0.615. \end{aligned}$$

Both values match `scikit-survival` exactly. The IPCW value is lower because both wrong pairs start from later deaths (C at 3 years, E at 5 years), whose weights are larger.

5.5 Single-variable baselines

Before fitting any model, each variable can be used directly as a risk score, with its direction fixed by clinical knowledge rather than learned from the outcome (missing values set to the median). Table 9 and Figure 10 give the results on the 3,173 training patients.

Table 9: Concordance of single-variable risk scores ($\tau = 7$ years).

Risk score	Harrell's C	IPCW C
Random numbers (reference)	0.509	0.509
+ bone-marrow blasts	0.631	0.616
− platelets	0.631	0.608
+ number of mutations	0.602	0.605
− haemoglobin	0.603	0.593
+ *TP53* mutated (0/1)	0.566	0.553
− *SF3B1* mutated (0/1)	0.554	0.544
+ monosomy 7 (0/1)	0.532	0.526
− bone-marrow blasts (wrong sign)	0.369	0.384

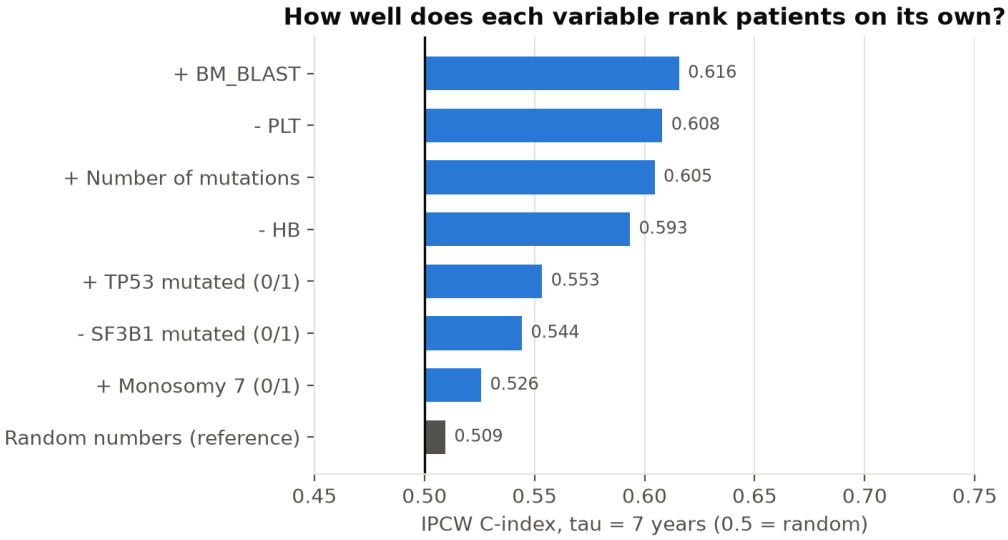


Figure 10: IPCW C-index of each single-variable risk score; 0.5 corresponds to random ranking.

Three lessons follow. First, reversing the sign of a score turns C into $1 - C$ ($0.631 \to 0.369$), so the direction of a risk score matters. Second, binary markers rank poorly on their own even when their log-rank test is overwhelming: *TP53* has $\chi^2 = 244.6$ but an IPCW C of only 0.553. Only 11.6% of patients carry the mutation, so most comparable pairs have equal scores and count one half. Statistical significance and ranking power are different things. Third, no single variable exceeds about 0.62, so combining them in a multivariable model is necessary.

6 Feature engineering (Stage 3)

Ninety-one features were built in five families (Table 10). Every feature is computed from one patient's own data, except two fixed choices that use only the predictors and never the outcome: the gene panel and the IPSS-R fill-in points. Missing clinical values are deliberately left empty at this stage, because the value used to fill them must be learned inside each cross-validation fold (Section 6.6).

Table 10: Feature families.

Family	Columns	Content
Clinical	6	blasts, haemoglobin, $\log(1+x)$ of 4 blood counts
Missing flags	6	1 if the clinical value was missing
Cytogenetic	18	abnormality count, 14 lesions, IPSS-R category, missing
Molecular summaries	15	counts, VAF, TP53, FLT3-ITD, MLL-PTD, 6 pathways
Gene indicators	45	1 if the gene is mutated
Clinical score	1	IPSS-R score
Total	91	

6.1 Clinical features

Blood counts are strongly right-skewed (Figure 4), so white cells, neutrophils, monocytes and platelets enter as $\log(1+x)$; the "1+" keeps recorded zeros finite. Blasts and haemoglobin are kept on their original scale. Each of the six variables also gets a 0/1 "was missing" flag.

6.2 Parsing the karyotype

A karyotype such as `46,xy,del(20)(q12)[2]/46,xy[18]` lists clones separated by "`/`", each with a chromosome count, the sex chromosomes, the abnormalities and a cell count in square brackets. Numerical changes are written `-7` (loss of chromosome 7) or `+8` (an extra chromosome 8); structural changes as `del`, `t`, `inv`, `der`, `add`, `i` and so on, and unidentified marker chromosomes as `mar`.

The text is irregular, especially in the test centre: upper and lower case, spaces, `{16}` or `[3/20]` cell counts, unbalanced brackets, and free text such as `Normal`, `complex` or `No mitosis`. The parser therefore searches for patterns instead of fully splitting the string:

1. lower-case the text, remove spaces, and convert `{}` to `[]`;
2. map free text: `Normal` to a normal karyotype, `complex` to at least three abnormalities, `>3 abnormalities` to four, and `No mitosis` or `Not evaluated` to missing;
3. delete cell counts, then collect the distinct numerical changes, the distinct structural changes and any marker chromosome across all clones;
4. do not count a translocation that only describes how a derivative chromosome was formed (`der(1)t(1;7)` is one abnormality), nor the common harmless variant `inv(9)(p12q13)`.

The parser was checked on 25 hand-written test strings covering these cases, and its monosomy 7 count (165 patients) agrees exactly with the simpler rule used in Stage 2. From the parsed karyotype the patient is assigned an IPSS-R cytogenetic category [9] (Table 11).

Table 11: IPSS-R cytogenetic categories as implemented.

Points	Category	Karyotype
0	Very good	$-$Y alone; del(11q) alone
1	Good	normal; del(5q), del(12p) or del(20q) alone; two including del(5q)
2	Intermediate	del(7q) alone, +8, +19, any other one or two abnormalities
3	Poor	-7; inv(3)/t(3;3)/del(3q); two including -7/del(7q); exactly three
4	Very poor	more than three abnormalities

Table 12 compares lesion frequencies. The test centre has nearly three times the training rate of complex karyotypes (31.3% against 10.9%) and of TP53 multi-hit status, defined as two or more TP53 mutations, a TP53 VAF of at least 0.5, or a TP53 mutation together with a 17p loss (18.9% against 6.7%).

Table 12: Lesion frequencies (% of patients with a usable karyotype; the first and last rows are % of all patients).

Lesion	Train	Test
Karyotype missing or unusable	11.0	10.1
Normal karyotype	58.2	38.2
Complex (≥ 3 abnormalities)	10.9	31.3
Monosomal karyotype	8.3	23.7
Monosomy 7	5.8	13.3
del(5q)	11.3	10.1
Trisomy 8	8.1	12.3
17p loss	2.1	7.4
TP53 multi-hit	6.7	18.9

6.3 Mutation features and the shared gene panel

The training file contains 124 genes and the test file 83, of which only 64 appear in both. Several genes frequent in training never appear in the test file, notably *KMT2C* (56 patients), *ZBTB33* (45), *EP300* (43) and *KMT2D* (32), which strongly suggests that the test centre used a smaller sequencing panel. If these genes were counted, "number of mutations" would mean something different in the two sets. All mutation features therefore use only the 64 shared genes; this removes 598 training mutations. A second mismatch was found: the test file never uses the label `splice_site_variant`, while 90 of its mutations have no effect label and are written in splice-site notation (`c.307+1G>A`); these were relabelled as splice-site variants.

The molecular features are: the number of mutations and of mutated genes, the maximum VAF, the number of truncating mutations (frameshift, stop-gained, splice-site), the number and maximum VAF of TP53 mutations, TP53 multi-hit status, FLT3 internal tandem duplication, MLL partial tandem duplication, six pathway counts (splicing, DNA methylation, chromatin, signalling, cohesin, transcription factors), and a 0/1 indicator for each of the 45 shared genes mutated in at least 1% (32) of training patients. Patients with no mutation receive zeros: "no mutation found" is information, not a missing value.

6.4 The IPSS-R score: a clinical benchmark

The revised International Prognostic Scoring System (IPSS-R) [8] is the standard clinical risk score for myelodysplastic syndromes, the dominant condition in this cohort (median blasts 3%). It adds points for five components:

Points	0	0.5	1	1.5	2–4
Cytogenetics	very good	–	good	–	2 int., 3 poor, 4 very poor
Blasts (%)	≤ 2	–	$(2, 5)$	–	2 if 5–10, 3 if > 10
Haemoglobin (g/dL)	≥ 10	–	8–<10	< 8	–
Platelets (G/L)	≥ 100	50–<100	< 50	–	–
ANC (G/L)	≥ 0.8	< 0.8	–	–	–

For example, a patient with a good karyotype (1 point), 7% blasts (2), haemoglobin 9.1 g/dL (1), platelets 45 G/L (1) and ANC 1.2 G/L (0) scores $1 + 2 + 1 + 1 + 0 = 5$, the "high" group.

A missing component receives the median points of the training patients (1 for cytogenetics, blasts and haemoglobin; 0 for platelets and ANC). This uses no outcome information. Groups are very low (≤ 1.5), low $(1.5, 3]$, intermediate $(3, 4.5]$, high $(4.5, 6]$ and very high (> 6).

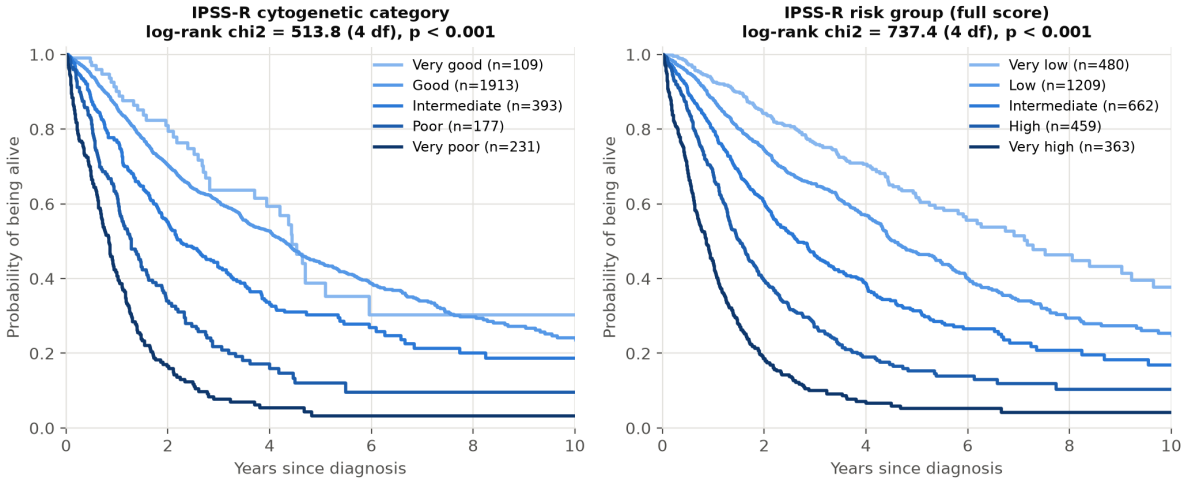


Figure 11: Kaplan–Meier curves by IPSS-R cytogenetic category (left) and by IPSS-R risk group (right) in the training data. Darker blue means higher risk.

Table 13: Survival by IPSS-R group in the training data.

Grouping	Group	n	Median (y)	$\hat{S}(3)$
Cytogenetic category	Very good	109	4.44	0.636
($\chi^2 = 513.8$)	Good	1913	4.28	0.606
	Intermediate	393	2.26	0.432
	Poor	177	1.28	0.217
	Very poor	231	0.84	0.076
Risk group	Very low	480	7.18	0.762
($\chi^2 = 737.4$)	Low	1209	4.56	0.654
	Intermediate	662	2.65	0.462
	High	459	1.52	0.270
	Very high	363	0.86	0.101

Both groupings separate survival cleanly and in the expected order (Figure 11, Table 13). The full score separates better than cytogenetics alone (χ^2 737.4 against 513.8), confirming that blasts and blood counts add information. Figure 12 applies the same score to the test centre: 33% of its patients are very high risk against 11% in training, and the median score is 5.0 against 3.0.

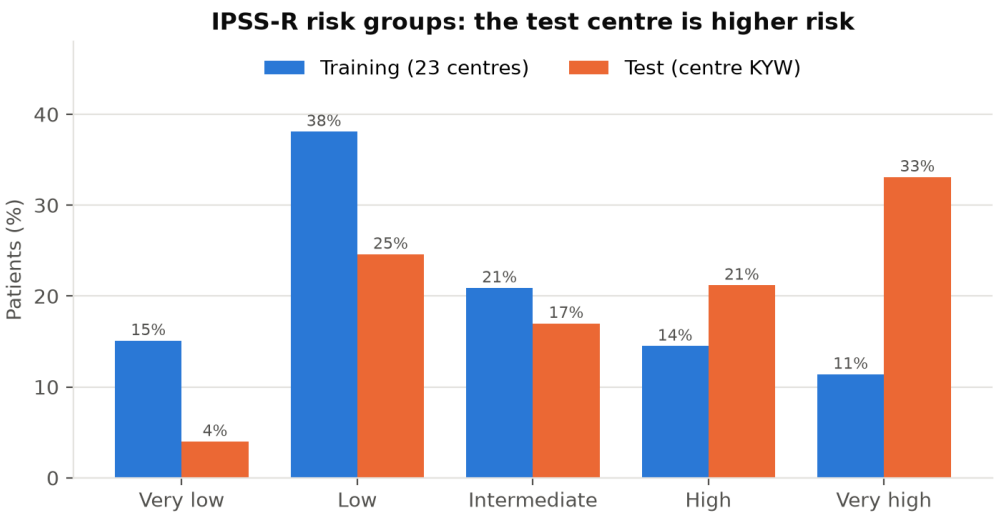


Figure 12: Share of patients in each IPSS-R risk group, training versus test.

6.5 Single-feature concordance

Figure 13 ranks features by their IPCW C-index when used alone (missing values filled with the median for this exploratory look only). The IPSS-R score is the strongest at 0.668, ahead of blasts (0.616) and the number of mutated genes (0.613). Platelets, haemoglobin, a normal karyotype and *SF3B1* mutation are protective ($C < 0.5$). Composite features such as counts and pathway sums rank better than single genes, for the tie-breaking reason found in Stage 2.

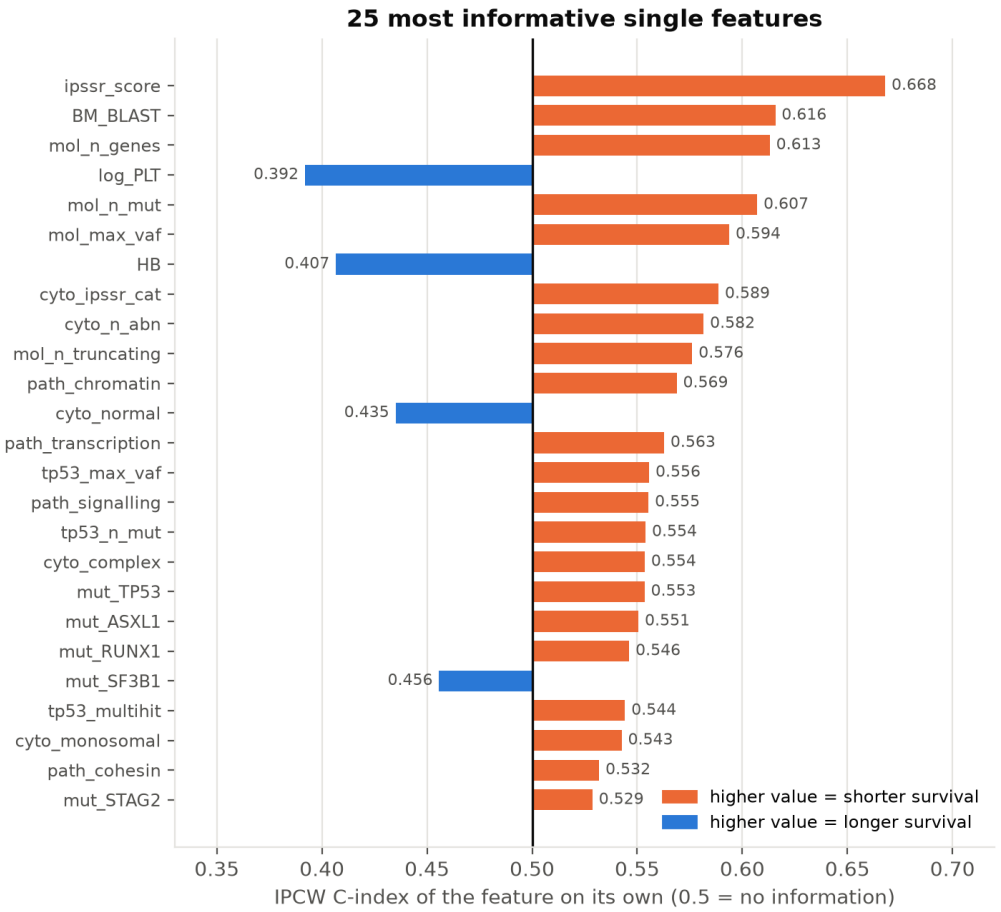


Figure 13: IPCW C-index of the 25 most informative features used alone. Orange: higher values go with shorter survival; blue: with longer survival.

6.6 Centre-grouped cross-validation

All patients of one centre are placed in the same fold, so each validation fold contains only centres unseen during training. The 23 centres are dealt into $K = 5$ folds like cards: they are put in a random order, and each in turn goes to the fold that currently has the fewest patients, which keeps the folds of similar size. Because there are only 23 centres, the split is repeated three times with different random orders, giving 15 validation folds. Two checks run in the code: no centre is split across folds, and the three repetitions are genuinely different partitions of the centres.

A bug found and fixed. The first version used `scikit-learn`'s `StratifiedGroupKFold` with `shuffle=True` and a different seed per repetition. The plot of fold scores in Stage 4 showed every value exactly three times: with only 23 groups, that splitter sorts the groups by their outcome profile before shuffling has any effect, so all three "repetitions" were the same five folds. The dealing procedure above replaced it, and the second check now guards against a repeat of the problem.

Table 14: Folds of the first repetition and the IPCW C-index of the (unfitted) IPSS-R score on each validation fold.

Fold	Patients	Centres	Death rate	Centre names	IPSS-R C_7
0	476	5	0.45	DUTH ICO MSK RMCN ROM	0.675
1	893	2	0.61	FUCE KI	0.659
2	478	6	0.38	CCH CGM FLO HIAE UOB VU	0.729
3	747	4	0.58	DUS GESMD HMS REL	0.629
4	579	6	0.40	IHBT MUV PV TUD UMG UOXF	0.679

The IPSS-R score is not fitted, so its variation across folds measures pure centre-to-centre noise. Over all 15 folds its IPCW C-index is 0.669 ± 0.029 (range 0.624–0.729), against 0.668 on all patients together. A difference in mean C-index of less than about 0.03 between two models is therefore not convincing by itself; the fair comparison is fold by fold (Section 7).

Filling missing values inside the folds. The median used to fill a missing value must be computed on the training part of each fold only. Otherwise the validation patients would help choose the fill-in value, a small leak of information from the data used to judge the model. Table 15 shows that these medians differ slightly between folds; Stage 4 will learn them inside each fold with a `scikit-learn` pipeline.

Table 15: Medians computed on the training part of each fold (repetition 0).

Fold	Blasts	Haemoglobin	$\log(1 + \text{PLT})$	$\log(1 + \text{monocytes})$
0	3.0	9.70	4.796	0.316
1	3.0	9.60	4.828	0.300
2	3.0	9.78	4.836	0.329
3	3.0	9.70	4.796	0.313
4	3.0	9.60	4.844	0.301
All patients	3.0	9.70	4.820	0.315

7 Cox regression (Stage 4)

7.1 The Cox proportional hazards model

The *hazard* $h(t)$ is the instantaneous risk of death at time t for someone still alive at t. The Cox model [10] writes the hazard of a patient with features $x = (x_1, \dots, x_p)$ as

$$h(t \mid x) = h_0(t)\, \exp(\beta_1 x_1 + \cdots + \beta_p x_p)\,, \tag{5}$$

where $h_0(t)$ is a baseline hazard shared by everyone and left completely unspecified. In plain language: every patient's risk curve has the same shape over time, scaled up or down by a patient-specific factor $\exp(\beta^\top x)$. The ratio of two patients' hazards therefore does not depend on t (the *proportional hazards* assumption), and e^{β_k} is the *hazard ratio* (HR) for a one-unit increase in x_k with the other features held fixed. The linear predictor $\eta = \beta^\top x$ is exactly the risk score the challenge asks for.

The coefficients are found by maximising the *partial likelihood*

$$L(\beta) = \prod_{i:\ \Delta_i = 1} \frac{\exp(\beta^\top x_i)}{\sum_{j:\ T_j \geq T_i} \exp(\beta^\top x_j)}. \tag{6}$$

Each factor asks: of all patients still at risk when patient i died, how likely was it to be patient i? Censored patients appear in the denominators while they are at risk, which is how the model uses them, and $h_0(t)$ cancels out.

Worked example. TP53 multi-hit has HR 3.21 in the model below, so its coefficient is $\ln 3.21 = 1.166$. Haemoglobin has HR 0.768 per standard deviation (2.0 g/dL), so $\beta = \ln 0.768 = -0.264$. A patient with TP53 multi-hit whose haemoglobin is one standard deviation *below* average, all else equal, has

$$\mathrm{HR} = \exp(1.166 - (-0.264)) = \exp(1.430) = 4.18 = 3.21 \times \frac{1}{0.768},$$

that is, about four times the instantaneous risk of death of an otherwise average patient.

7.2 Penalised Cox: the elastic net

With 84 features, some rare, ordinary maximisation of Equation (6) over-fits. The elastic net [11, 12] instead maximises

$$\ln L(\beta) \; - \; \alpha \left(\rho \sum_k |\beta_k| + \frac{1-\rho}{2} \sum_k \beta_k^2 \right), \tag{7}$$

where $\alpha > 0$ sets the overall strength of the penalty and $\rho \in (0, 1]$ the mix: $\rho = 1$ is the lasso, which sets many coefficients exactly to zero, and small ρ is close to ridge regression, which shrinks all coefficients towards zero without removing them. Features are standardised first so the penalty treats them equally.

Nested cross-validation. α and ρ must be chosen without looking at the validation fold. Inside each outer training set, an inner four-fold split, also grouped by centre, scores 30 values of α for each $\rho \in \{0.1, 0.5, 1\}$; the best pair is refitted on the whole outer training set and only then applied to the outer validation fold. Missing values are filled with medians and features are standardised inside a `scikit-learn` pipeline, so these steps are also learned only on the data the model is trained on.

7.3 Models compared

All models are evaluated on the same 15 centre-grouped validation folds, with the censoring curve $\hat{G}$ of the metric estimated on each fold's training part:

A. IPSS-R score the published score, no fitting (reference).

B. Benchmark-style Cox blasts, haemoglobin, platelets and the number of mutations. The challenge's benchmark notebook is available only to registered participants; this is my reconstruction from the challenge description and may differ in detail.

C. IPSS-M-style Cox the five blood values, the IPSS-R cytogenetic category, a flag for a missing karyotype, the 16 "main effect" genes of the IPSS-M molecular score [17] (including TP53 multi-hit and MLL-PTD), and the number of mutated IPSS-M "residual" genes capped at 2. These 24 variables come from the literature, not from this data. A very small ridge penalty (0.1) keeps the fit stable.

D. Elastic-net Cox all 84 features except monocytes and the missing flags.

E. Elastic-net Cox + D plus log monocytes and the six missing flags.

7.4 Cross-validated results

Table 16: IPCW C-index ($\tau = 7$) over 15 centre-grouped validation folds. "Paired" is the mean over folds of (model C − IPSS-R C on the same fold).

Model	Mean	SD	Range	Harrell's C	Paired	Folds won
A. IPSS-R score	0.669	0.029	0.624–0.729	0.692	–	–
B. Benchmark-style Cox	0.688	0.029	0.657–0.756	0.703	+0.019	13/15
C. IPSS-M-style Cox	0.716	0.023	0.686–0.765	0.736	+0.047	15/15
D. Elastic-net Cox	**0.722**	0.021	0.696–0.769	0.740	+0.053	15/15
E. Elastic-net Cox +	0.721	0.021	0.696–0.769	0.739	+0.053	15/15

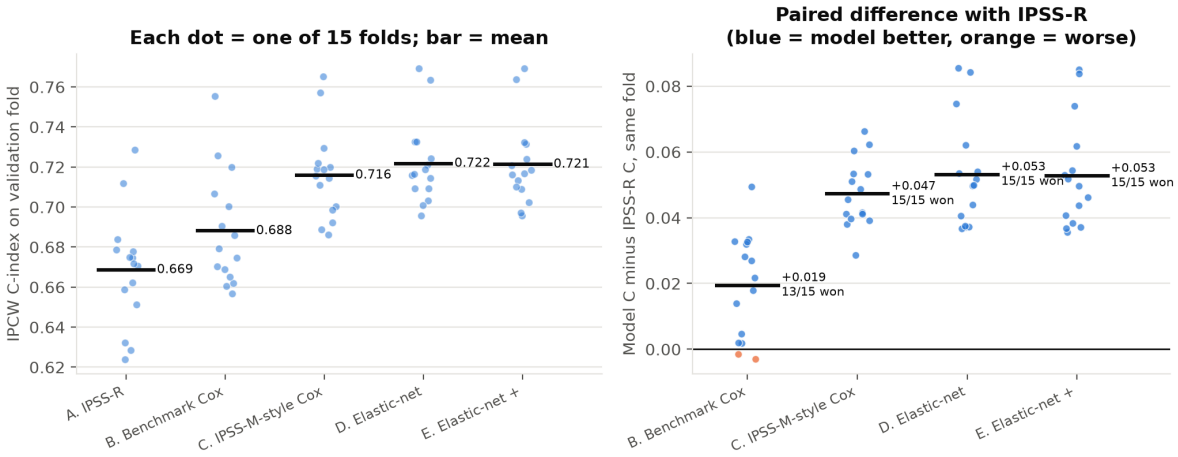


Figure 14: Left: IPCW C-index of each model on each of the 15 validation folds. Right: the paired difference with the IPSS-R score on the same fold.

Table 17: Further paired comparisons over the 15 folds.

Comparison	Question	Mean diff.	SD	Folds won
C − B	clinical feature set vs benchmark	+0.028	0.014	14/15
D − C	84 penalised features vs 24 chosen	+0.006	0.009	10/15
E − D	adding monocytes and missing flags	−0.0004	0.001	4/15

Four conclusions follow from Tables 16 and 17 and Figure 14:

1. Every fitted Cox model beats the unfitted IPSS-R. The gains of C, D and E are convincing: about +0.05, larger than the fold-to-fold spread, and won on all 15 folds.

2. The real step forward is the feature set. Moving from the four benchmark variables to the 24 literature-chosen variables adds +0.028 and wins 14 of 15 folds.

3. Letting a penalised model use all 84 features adds only +0.006 beyond those 24 variables (10 of 15 folds), a small and uncertain gain.

4. Monocytes and the missing-value flags add nothing and lose 11 of 15 folds. This settles the Stage 1 concern: the variable missing for 74% of test patients is safely dropped.

On the training data itself, the IPSS-M-style model reaches a concordance of 0.743 against 0.716 in cross-validation: the gap is the optimism of judging a model on the patients it was fitted to.

7.5 Hazard ratios

Figure 15 shows the hazard ratios of the IPSS-M-style model refitted on all 3,173 patients (missing values filled with the median; blood values per standard deviation, the cytogenetic category and residual-gene count per step, and genes as mutated versus not).

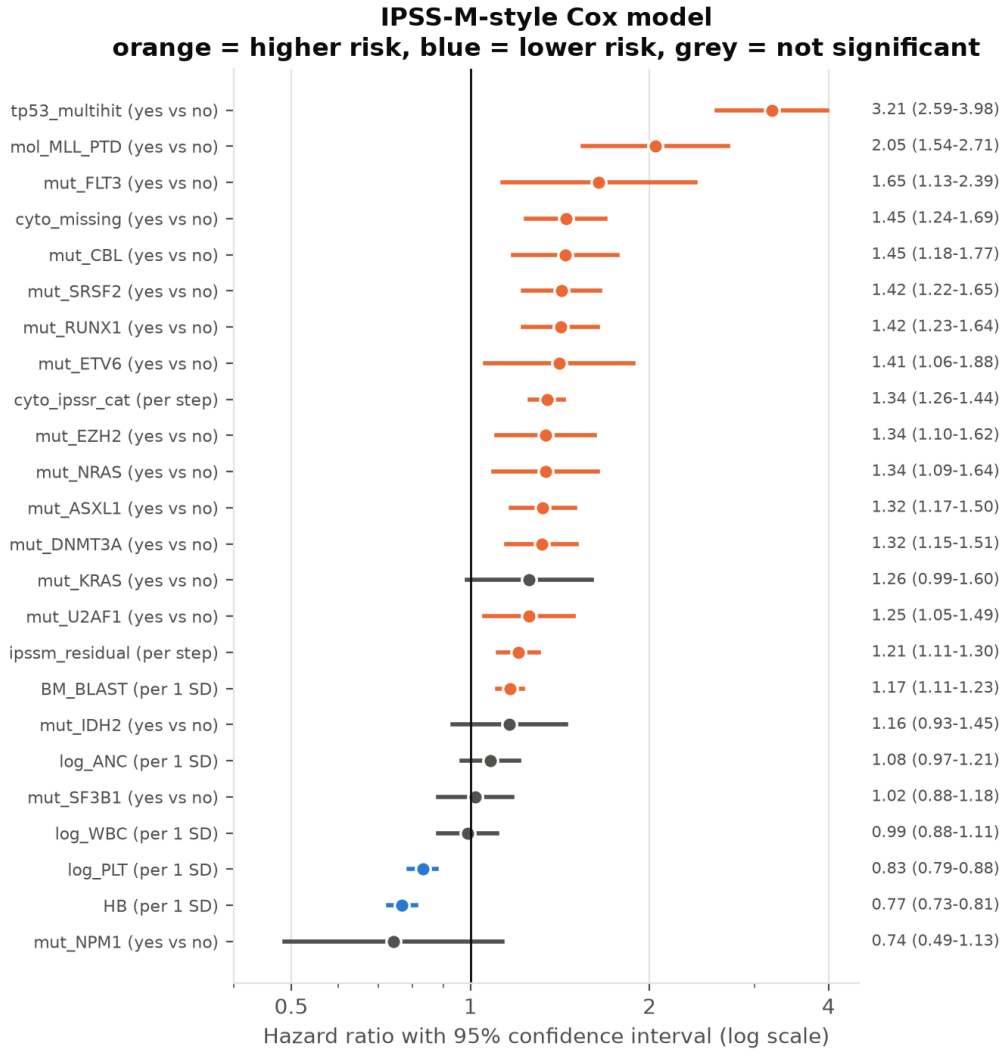


Figure 15: Hazard ratios with 95% confidence intervals of the IPSS-M-style Cox model. Orange: significantly higher risk; blue: significantly lower risk; grey: interval includes 1.

TP53 multi-hit is by far the strongest factor (HR 3.21, 95% CI 2.59–3.98), followed by MLL partial tandem duplication (2.05) and FLT3 mutation (1.65). Each step up the IPSS-R cytogenetic scale multiplies the hazard by 1.34, and a missing karyotype by 1.45, perhaps because failed karyotypes are more common in sicker patients. Higher haemoglobin (HR 0.77 per SD) and platelets (0.83 per SD) are protective. Two results differ from Stage 2. First, *SF3B1* is no longer protective (HR 1.02, CI 0.88–1.18): its favourable Kaplan–Meier curve is explained by the other variables, since *SF3B1*-mutated patients tend to have low blasts and a good karyotype. Second, white cells and neutrophils add nothing once the other variables are known.

7.6 Checking proportional hazards

If the proportional-hazards assumption holds, each coefficient β_k is constant over time. Grambsch and Therneau [13] showed that the scaled Schoenfeld residual of a death at time t, plus $\hat{\beta}_k$, estimates $\beta_k(t)$; a trend over time signals a violation. Table 18 gives the test for the largest statistics.

Table 18: Proportional-hazards test (rank-transformed time), largest statistics.

Variable	Test statistic	p
NPM1 mutated	23.10	1.5×10^{-6}
Karyotype missing	17.07	3.6×10^{-5}
log platelets	16.04	6.2×10^{-5}
SF3B1 mutated	5.12	0.024
U2AF1 mutated	3.64	0.056

Four of 24 variables have $p < 0.05$. With 1,600 deaths even small departures become significant, so Figure 16 shows the size of the two clearest ones. *NPM1* is harmful in the first months ($\beta(t) \approx +2$) and strongly protective afterwards ($\beta(t) \approx -2.5$ beyond four years): its single average coefficient of -0.30 hides two opposite effects. The protection of a high platelet count is concentrated in the first year and fades to almost nothing after three years. The other 20 variables show no evidence of change. Because the metric only ranks patients, mild violations matter less than they would for predicting survival probabilities, but they motivate the tree-based models of Stage 5, which do not assume proportional hazards.

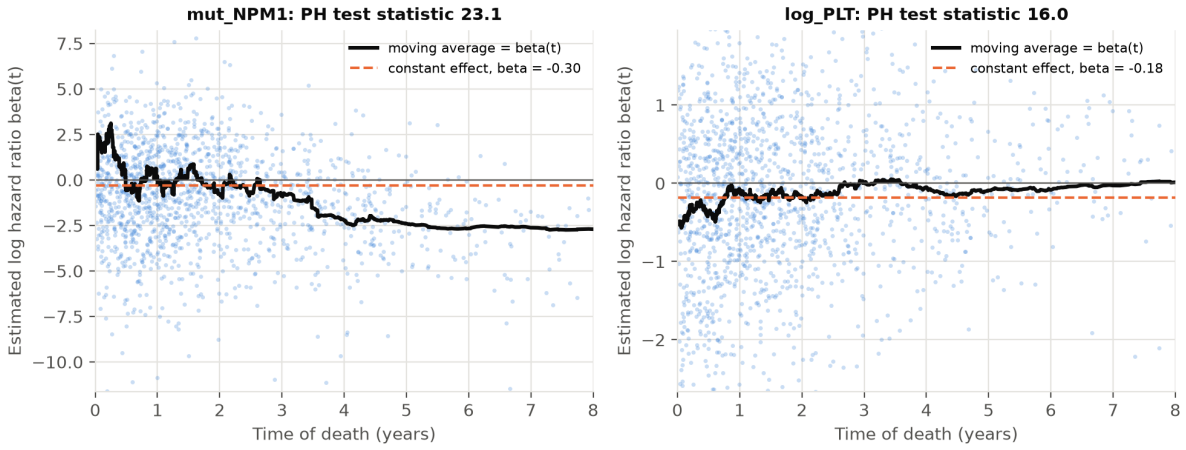


Figure 16: Estimated time-varying effect $\beta(t)$ (scaled Schoenfeld residual plus $\hat{\beta}$; black line: moving average of 150 deaths) against the constant effect assumed by the Cox model (dashed).

7.7 The final elastic-net model

Refitted on all training patients with the same nested procedure (inner five-fold centre-grouped split), the elastic net chooses $\rho = 0.1$, close to ridge, and $\alpha = 0.190$, keeping 44 of 84 coefficients non-zero (Figure 17). The largest effects are the IPSS-R score itself, the number of mutated genes, the TP53 and overall maximum VAF, and, on the protective side, haemoglobin and platelets.

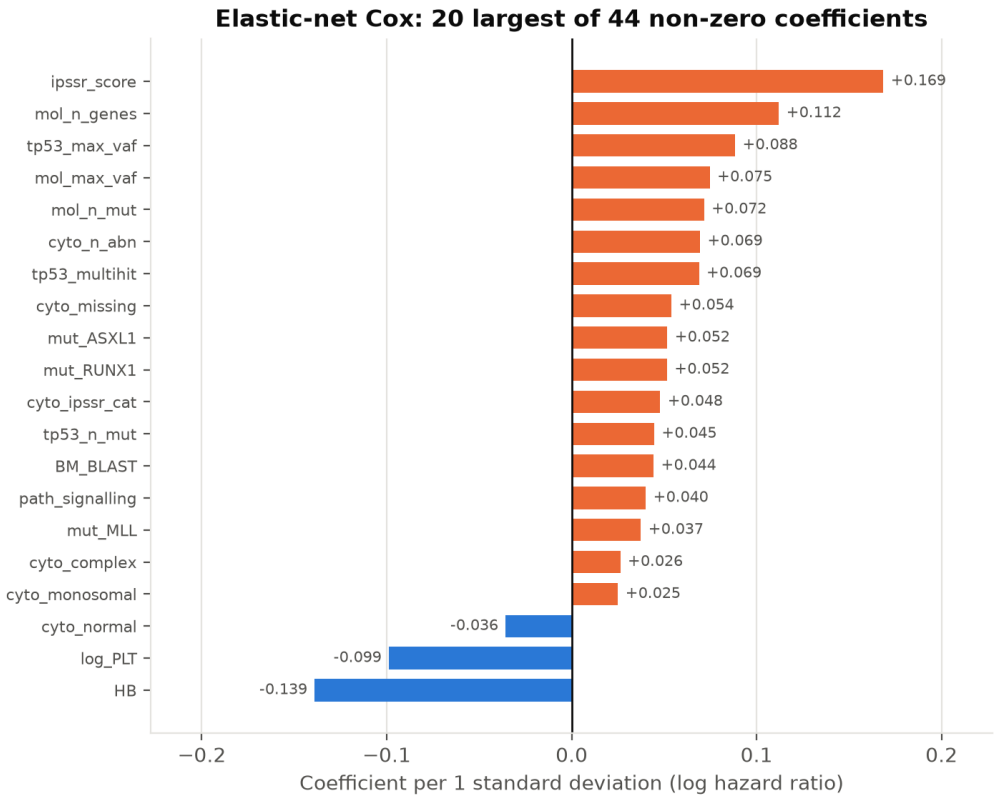


Figure 17: The 20 largest coefficients of the final elastic-net Cox model, per standard deviation of each feature.

To see what a C-index of 0.72 means in practice, each training patient was given the risk score of the elastic-net model trained *without their centre* (first repetition), and patients were split into quartiles within each fold. Figure 18 and Table 19 show the result.

Table 19: Survival by out-of-fold risk quartile (elastic-net Cox, unseen centres).

Quartile	n	Median (y)	$\hat{S}(3)$	$\hat{S}(7)$
Q1 (lowest risk)	795	8.28	0.834	0.564
Q2	793	4.40	0.643	0.326
Q3	791	2.42	0.431	0.157
Q4 (highest risk)	794	1.07	0.130	0.057

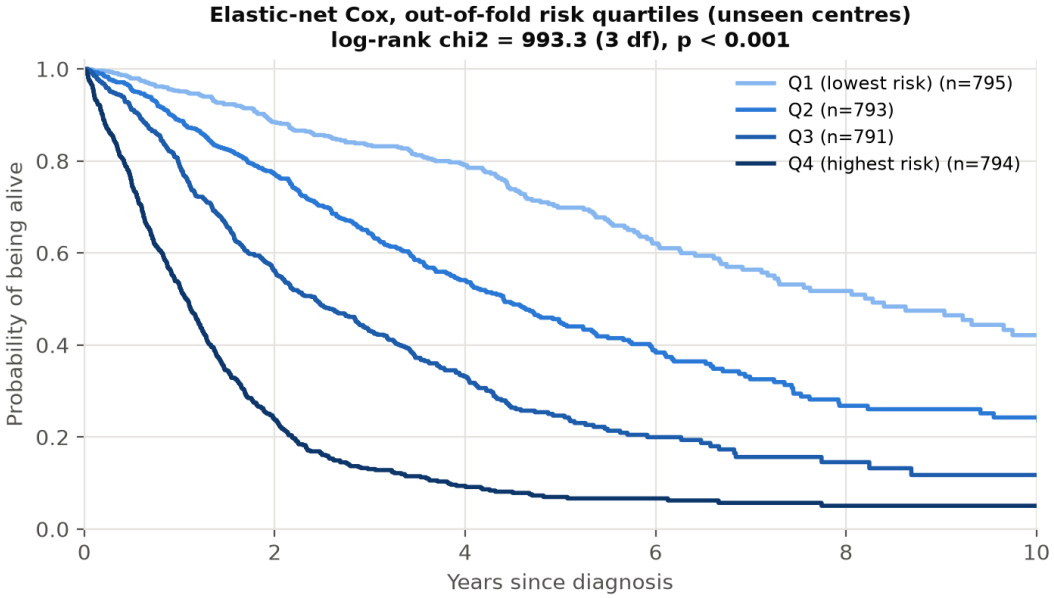


Figure 18: Kaplan–Meier curves by out-of-fold risk quartile; log-rank $\chi^2 = 993.3$ (3 df).

The median survival falls from 8.3 years in the lowest-risk quarter to 1.1 years in the highest, and three-year survival from 83% to 13%. For comparison, the five IPSS-R groups gave $\chi^2 = 737.4$ with four degrees of freedom (Stage 3).

7.8 Submissions

Three submission files were produced from models fitted on all 3,173 training patients: the benchmark-style Cox, the IPSS-M-style Cox and the elastic-net Cox. Each contains the 1,193 test IDs and a `risk_score`. As a sanity check, the elastic-net mean risk score is +0.43 on the test centre against 0 on the training data, consistent with the sicker test population. Two files were submitted (Table 20).

Table 20: Public leaderboard scores of the Stage 4 submissions (IPCW C-index on the public part of the test centre).

Model	Cross-validation	Public score
D. Elastic-net Cox (84 features)	0.722	**0.7543**
C. IPSS-M-style Cox (24 variables)	0.716	0.7374
Difference D − C	+0.006	+0.017

Two observations. First, both public scores are *higher* than cross-validation, the reverse of the usual drop on a new centre. A plausible reason is the case mix found in Stage 3: the test centre spreads its patients over a wider range of risk (a third are IPSS-R very high, yet low-risk patients remain), and a population with a wider range of risk is easier to rank, so the C-index rises. The leaderboard score therefore reflects the centre as much as the model, and models will continue to be chosen by paired cross-validation rather than by public score.

Second, the gap between the two models almost triples at the test centre, from +0.006 in cross-validation to +0.017. Both scores are measured on the same public patients, so the comparison is paired. A likely explanation is that the elastic-net's extra features describe the high-risk end in detail (number of abnormalities, monosomal karyotype, TP53 VAF, number of mutated genes), whereas model C summarises it coarsely (the IPSS-R cytogenetic category

stops at "more than three abnormalities"). The test centre has many more such patients than the training centres. A lesson for the write-up: cross-validation on centres unlike the target can understate the value of features that matter most for the target.

8 Machine-learning survival models (Stage 5)

Cox models assume that each feature acts the same way at all times (proportional hazards) and adds to the log hazard without interacting with the others. Tree-based models drop both assumptions. Two were fitted on the same 84 features as the elastic-net and evaluated on the same 15 centre-grouped folds.

8.1 Random survival forest

A random survival forest [14] grows many survival trees. Each tree is grown on a bootstrap sample of patients (drawn with replacement) and, at every split, considers only a random subset of the features. A split is chosen to separate the survival curves of the two child nodes as much as possible (by the log-rank statistic of Stage 2). Each leaf stores the Nelson–Aalen estimate of the cumulative hazard of its patients, and the forest averages these over trees. The risk score is the predicted cumulative hazard summed over the observed event times: more expected deaths means higher risk. Settings: 300 trees, at least 15 patients per leaf, and $\sqrt{84} \approx 9$ features tried per split. These were fixed rather than tuned, because forests are not very sensitive to them and one fit takes about six seconds.

8.2 Gradient boosting with a Cox loss

Gradient boosting [15] builds the risk score as a sum of small trees,

$$\eta(x) = \nu \sum_{m=1}^{M} f_m(x),$$

added one at a time. Each new tree f_m is fitted to the gradient of the negative Cox log partial likelihood (Equation (6)) at the current scores. In plain language, each tree pushes the scores in the direction that most improves the Cox fit, and the learning rate $\nu = 0.03$ keeps every step small. The XGBoost implementation [16] was used, with 80% of patients and half the features sampled for each tree and a minimum leaf weight of 10. It handles missing values itself by learning which branch they should follow. The tree depth (2, 3 or 4) and the number of trees (50 to 600) were chosen by the same nested, centre-grouped inner cross-validation as the elastic-net. In the first run, the smallest number of trees offered (100) was chosen in 7 of 15 folds. A choice at the edge of the grid means the best value might lie beyond it, so 50 and 150 were added. After that, 50 was never chosen, the choices spread between 100 and 300 trees, and the results did not change.

8.3 Results

Table 21: IPCW C-index over the same 15 centre-grouped folds; paired differences are with the elastic-net Cox (D).

Model	Mean	SD	Range	Paired vs D	Folds won vs D
A. IPSS-R score	0.669	0.029	0.624–0.729	−0.053	–
D. Elastic-net Cox	0.722	0.021	0.696–0.769	–	–
F. Random survival forest	0.719	0.021	0.694–0.774	−0.003	4/15
G. XGBoost Cox	**0.723**	0.021	0.698–0.767	+0.001	10/15

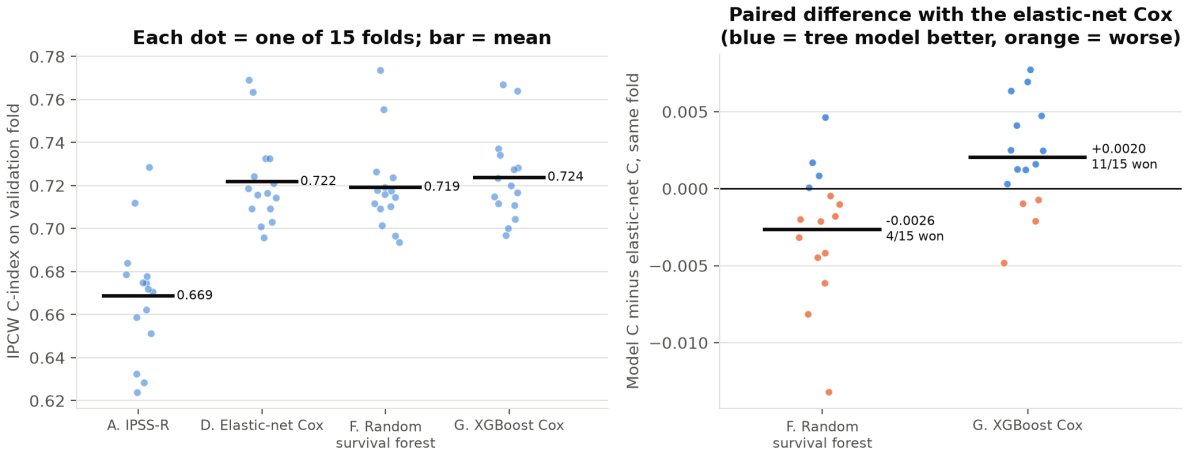


Figure 19: Left: IPCW C-index on each validation fold. Right: paired difference with the elastic-net Cox on the same fold.

The tree models do *not* beat the elastic-net (Table 21, Figure 19). XGBoost ties it (+0.001, 10 of 15 folds, a paired SD of 0.003), and the forest is slightly worse (−0.003, 4 of 15). This is a genuine result, not a failure of tuning: the inner cross-validation consistently preferred small models (100 to 300 shallow trees at a low learning rate), which signals that there is little structure left for trees to find beyond what the features already encode. Two reasons are likely. The engineered features already contain the main non-linearities and interactions, since the IPSS-R score, the cytogenetic categories and TP53 multi-hit status are themselves expert-designed combinations. And with about 1,600 deaths, the data are too small to estimate reliably the subtle time-varying effects found in Stage 4.

Accuracy by horizon. Figure 20 recomputes the out-of-fold IPCW C-index with horizons τ from one to seven years. All models rank early deaths better than late ones, falling from about 0.76 at one year to 0.72 at seven. The only visible advantage of XGBoost is at one year (0.760 against 0.756 for the elastic-net), consistent with trees capturing some short-term effects such as the early harm of *NPM1* mutation. The fitted models gain most over the IPSS-R at long horizons (0.051 at seven years against 0.041 at one), because the IPSS-R was designed for myelodysplastic syndromes rather than for this mixed cohort.

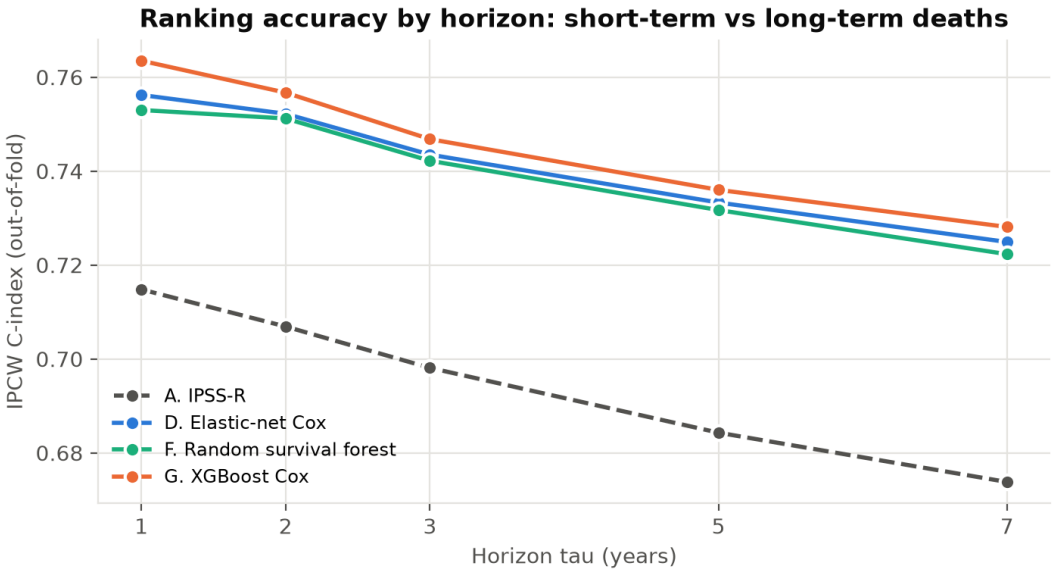


Figure 20: Out-of-fold IPCW C-index (repetition 0, mean of five folds) for horizons τ of one to seven years.

What the boosted trees use. Figure 21 shows each feature's share of the total improvement in the Cox loss ("gain") in the final XGBoost model. The IPSS-R score (20%) and the number of mutated genes (12%) dominate, followed by haemoglobin, TP53 VAF and platelets: the same picture as the elastic-net coefficients (Figure 17). Gain measures how much a model *uses* a feature, not its causal effect, and correlated features share the credit.

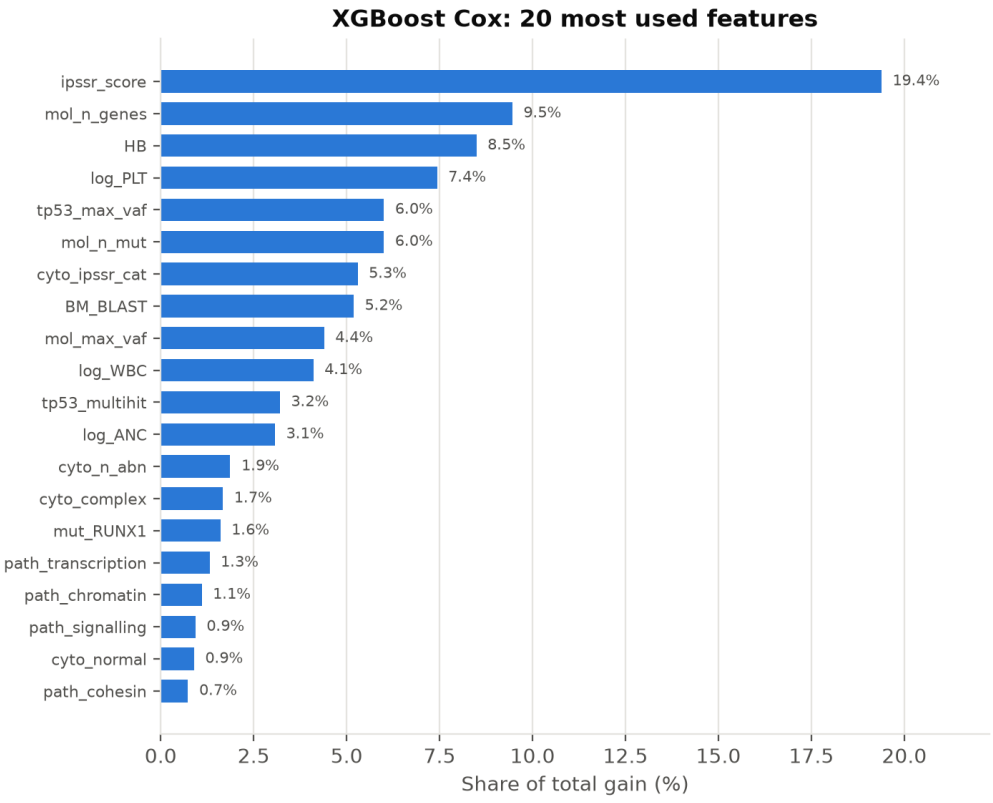


Figure 21: XGBoost Cox: the 20 features with the largest share of total gain.

How similar are the models? On the test centre, the Spearman rank correlation between the risk scores of the elastic-net and XGBoost is 0.960, between the elastic-net and the forest 0.967, and between XGBoost and the forest 0.969. The three models largely agree on who is at high risk, which limits how much averaging them can help.

9 Next steps

Stage 6 will combine the models in a rank-averaged ensemble, evaluated on the same folds, and examine permutation importance on held-out folds, then choose the final submission by paired cross-validation. Given the high correlation between the models, a modest gain is expected.

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